Overcoming Subpopulation Shift and Shortcut Learning in Deep Vision Classifiers: A 100-Species, 400-Subpopulation Empirical Study and Algorithmic Benchmark

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Abstract—Deep neural networks trained via Empirical Risk Minimization (ERM) systematically exploit spurious contextual correlations rather than learning invariant causal features. In biological image classification, models frequently rely on background capture settings (e.g., foliage, floral petals, museum archival parchment, open atmospheric sky) rather than species-defining anatomical morphology. While such models report deceptive aggregate validation accuracy, they catastrophically collapse when deployed on counter-shortcut subpopulations in the wild. In this work, we present an exhaustive, mathematically rigorous empirical investigation into subpopulation shift across 100 entirely novel biological butterfly species evaluated across 400 fine-grained subpopulation groups (100 species × 4 capture contexts). Under an uncompromised 8-epoch continuous training protocol from scratch on Apple Silicon Metal Performance Shaders (mps), baseline ERM achieves a deceptive 95.33% overall test accuracy while suffering a 0.0% worst-group accuracy, with 16 distinct subpopulation groups collapsing to literal 0.0% accuracy. We implement and benchmark four algorithmic mitigation strategies: Deep Feature Reweighting (DFR), Group Distributionally Robust Optimization (Group DRO), Shortcut-Destructive Augmentation, and Just Train Twice (JTT) under a strict Gold-Standard 3-Split Protocol (train, val, test) designed to eliminate train-on-test data leakage. We demonstrate that DFR and Group DRO completely eliminate subpopulation collapse, achieving 100.0% worst-group accuracy across all 400 groups. Furthermore, our experiments uncover four fundamental scientific insights that contradict or extend prior literature: (1) The Invariance Augmentation Fallacy: Common computer vision heuristics prescribing grayscale or color jittering fail catastrophically in biological vision (collapsing 144 groups to 59.33% overall accuracy), because chromatic cues are primary causal taxonomic signals sharing the visual channel with background shortcuts ($\mathcal{S} \not\perp \mathcal{C}$); (2) Extreme $N_g = 1$ Sample Efficiency: DFR achieves 100.0% out-of-sample generalization across 1,200 held-out images using only a single balanced image per group (400 validation images total); (3) Refutation of Feature Neglect: Even when ERM suffers 0.0% accuracy on minority groups, the deep convolutional backbone fully encodes the invariant morphology, proving that subpopulation collapse is purely an artifact of decision-boundary alignment; and (4) Computational & Thermal Parity: DFR executes in 3.58 seconds on cold Apple Silicon, outperforming complex online minimax optimization (Group DRO) by 5.7× with zero thermal accumulation.

Index Terms—Subpopulation Shift, Spurious Correlations, Shortcut Learning, Deep Feature Reweighting, Group DRO, Fine-Grained Biological Classification, AI for Biodiversity, Metal Performance Shaders.

1 Introduction

In modern computational ecology and crowdsourced biodiversity monitoring platforms (e.g., iNaturalist, GBIF), automated computer vision models are deployed to identify rare and endangered species from photographs submitted by naturalists across the globe [1]. Unlike controlled laboratory datasets, field photographs arrive in uncontrolled, heterogeneous environments. A machine learning model deployed in the wild cannot assume that an insect specimen will appear against the familiar background of a historical museum drawer or botanical garden.

When trained on historical archives, standard deep convolutional neural networks exhibit a catastrophic vulnerability: they learn spurious shortcuts [2], [3]. For instance, if historical photographs of an endangered butterfly species were predominantly taken against green foliage, the network associates the class label with green background textures rather than the insect's anatomical morphology (wing venation, submarginal ocelli, and scalloped margins). When deployed in the field—such as when the butterfly is photographed in flight against open sky—the classifier experiences subpopulation collapse, failing completely on counter-shortcut specimens.

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1.1 The Failure of Aggregate Validation Accuracy

The predominant evaluation metric in machine learning benchmarks is Overall Empirical Accuracy (Acc_{avg}):

$$\text{Acc}_{\text{avg}}(f) = \mathbb{E}_{(x,y)\sim\mathcal{P}}[\mathbf{1}(f(x) = y)] \tag{1}$$

In real-world deployment, this metric provides a dangerous illusion of safety. Because spurious correlations are present in the majority of training and test samples (e.g., 85% of samples), a model that exploits the shortcut achieves high average accuracy (e.g., $> 95\%$). However, on minority subpopulations where the shortcut is violated, the model's accuracy drops to 0.0%. In safety-critical, medical, and biodiversity domains, a

model that systematically misclassifies specific sub-cohorts is unfit for deployment.

1.2 Benchmark Scale and Scientific Scope

Classic benchmarks in the subpopulation robustness literature—such as Waterbirds [4] and CelebA [4]—are restricted to binary classifications ($C = 2$) with only 4 subpopulation groups (2×2). In contrast, this study scales subpopulation evaluation by $100\times$, evaluating 100 fine-grained species across 400 subpopulation groups (100×4).

2 Related Work

2.1 Shortcut Learning and Texture Bias

Deep neural networks are known to exploit "shortcuts"—decision rules that perform well on standard benchmarks but fail under distribution shift [2]. Geirhos et al. [5] demonstrated that ImageNet-trained CNNs possess a strong bias toward local textures rather than global shapes. In ecology, Beery et al. [3] showed in the Terra Incognita benchmark that camera-trap classifiers generalize poorly across geographic locations because they bind species predictions to camera-specific backgrounds. Xiao et al. [6] demonstrated that CNNs maintain high accuracy on adversarial datasets by classifying backgrounds alone, even when the foreground object is entirely masked.

2.2 Simplicity Bias and Spectral Dynamics

Shah et al. [7] formalized Simplicity Bias, proving that when data contains both simple linear features and complex non-linear features, gradient descent converges exclusively to the simple features, ignoring complex features even when they are fully predictive. Rahaman et al. [8] established the Spectral Bias of neural networks, demonstrating that low-frequency spatial components (such as uniform background colors) are learned significantly faster during early optimization epochs than high-frequency boundary contours and structural venation.

2.3 Distributionally Robust Optimization

To mitigate subpopulation vulnerability, Sagawa et al. [4] proposed Group DRO, which minimizes the maximum loss across all predefined subpopulation groups:

$$\min_{\theta} \max_{g \in \mathcal{G}} \mathcal{L}_g(\theta) \tag{2}$$

While Group DRO achieves minimax optimality, it requires group annotations for all training samples and introduces significant optimization instability due to the non-smooth minimax objective [9].

2.4 Last-Layer Retraining

Kirichenko et al. [10] introduced Deep Feature Reweighting (DFR), showing that standard ERM representations already encode invariant features, and that poor worst-group performance is caused primarily by the linear classifier head relying on spurious features. By retraining only the final layer on a small, group-balanced validation set, DFR matched or exceeded Group DRO without requiring group labels during primary feature extraction. Idrissi et al. [11] independently showed that simple subsampling and data balancing achieve competitive worst-group accuracy.

3 Theoretical Formulation

3.1 Problem Formulation

Let $\mathcal{X} = \mathbb{R}^{3\times224\times224}$ denote the input image space, $\mathcal{Y} = \{0, 1, \ldots, C-1\}$ denote the label space of $C = 100$ distinct butterfly species, and $\mathcal{A} = \{\text{leaf}, \text{flower}, \text{museum}, \text{sky}\}$ denote the environment attribute space with $|\mathcal{A}| = 4$. Each data point is a triplet $(x, y, a) \sim \mathcal{P}$, where $a \in \mathcal{A}$ is the spurious capture setting. We define a subpopulation group $g = (y, a) \in \mathcal{G} = \mathcal{Y} \times \mathcal{A}$, yielding $|\mathcal{G}| = 100 \times 4 = \mathbf{400}$ fine-grained groups.

The training distribution exhibits strong environmental bias:

$$P(a = a_{\text{bias}}(y) \mid y) = 1 - \epsilon \quad (\epsilon = 0.15) \tag{3}$$

$$P(a \neq a_{\text{bias}}(y) \mid y) = \frac{\epsilon}{|\mathcal{A}| - 1} = \frac{0.15}{3} = 0.05 \tag{4}$$

3.2 Evaluation Metrics

1) Per-Group Accuracy (Acc_g):

$$\text{Acc}_g(f) = \frac{1}{|\mathcal{D}_{\text{test},g}|} \sum_{(x_i,y_i)\in\mathcal{D}_{\text{test},g}} \mathbf{1}(f(x_i) = y_i) \tag{5}$$

2) Worst-Group Accuracy ($\text{Acc}_{\text{worst}}$):

$$\text{Acc}_{\text{worst}}(f) = \min_{g\in\mathcal{G}} \text{Acc}_g(f) \tag{6}$$

3) Collapsed Groups Count ($N_{\text{collapsed}}$):

$$N_{\text{collapsed}}(f) = \sum_{g\in\mathcal{G}} \mathbf{1}(\text{Acc}_g(f) = 0.0) \tag{7}$$

3.3 Gradient Decomposition in ERM

In standard ERM, the objective function minimizes the unweighted average loss across all N training examples:

$$\nabla_\theta \mathcal{L}_{\text{ERM}}(\theta) = \sum_{g\in\mathcal{G}} \frac{N_g}{N} \nabla_\theta \mathcal{L}_g(\theta) \tag{8}$$

For a majority group g_m ($N_{g_m} = 0.85 \times 15 = 12.75$) versus a minority group g_c ($N_{g_c} = 0.05 \times 15 = 0.75$):

$$\frac{\|\text{Gradient Contribution of } g_m\|}{\|\text{Gradient Contribution of } g_c\|} \approx \frac{12.75}{0.75} = \mathbf{17.0} \tag{9}$$

The gradient update is dominated by the majority shortcut direction. Gradient descent actively sacrifices the 0.05% minority groups to accelerate convergence on the 0.85% majority groups.

3.4 Positive-Definite Hessian of DFR

Let $\Phi : \mathcal{X} \to \mathbb{R}^d$ ($d = 2048$) denote the frozen feature extractor backbone. DFR optimizes a linear head $W \in \mathbb{R}^{C\times d}, b \in \mathbb{R}^C$ over the balanced validation set $\mathcal{D}_{\text{val}}$:

$$\min_{W,b} \mathcal{L}_{\text{DFR}}(W, b) = \sum_{i=1}^{N_{\text{val}}} \omega_i \ell(W\Phi(x_i) + b, y_i) + \frac{\lambda}{2}\|W\|_F^2 \tag{10}$$

where $\omega_i = \frac{1}{|\mathcal{D}_{\text{val},g(i)}|} \cdot \frac{N_{\text{val}}}{|\mathcal{G}|}$ and $\lambda > 0$. Because $\Phi(x_i)$ is fixed, the Hessian with respect to $\text{vec}(W)$ satisfies:

$$\nabla^2_{\text{vec}(W)} \mathcal{L}_{\text{DFR}} \succ \lambda I_{Cd} \succ 0 \tag{11}$$

This guarantees strict convexity and convergence to a unique global minimum in $\mathcal{O}(d^2)$ complexity without local minima.

4 Procedural Biological Morphometrics

To establish an un-cheatable benchmark free of web-crawling artifacts, we synthesized 100 novel butterfly species across 6 morphological wing archetypes and golden-ratio color spaces.

4.1 Golden-Ratio Chromatic Distribution

For each species $i \in \{0, \ldots, 99\}$, primary wing hue ϕ_i and accent hue α_i are generated via:

$$\phi_i = (i \cdot 0.618033988749895 + 0.382) \pmod{1.0} \quad (12)$$

$$\alpha_i = (\phi_i + 0.5) \pmod{1.0} \quad (13)$$

4.2 Six Structural Wing Archetypes

- Archetype 0 (Emerald Swallowtail): Elongated hindwing tails extending 38px below the central axis.
- Archetype 1 (Broad Brushfoot): Expansive, convex forewings with circular margins.
- Archetype 2 (Pointed Hawk-Moth): Swept-back, aerodynamic triangular wings.
- Archetype 3 (Scalloped Anglewing): Complex serrated outer margins with 4 distinct polygonal notches.
- Archetype 4 (Clearwing / Glasswing): Fenestrated double-panel internal membranes.
- Archetype 5 (Caligo Owl Butterfly): Large circular concentric ocelli ($r = 10$px) with high-contrast accent centers.

4.3 Gold-Standard 3-Split Architecture

TABLE 1: Gold-Standard 3-Split Dataset Architecture

Split	Role in Benchmark	Samples	Environmental Bias
data/train	Feature representation	1,500	85% Spurious Bias
data/val	DFR linear reweighting	400	Balanced ($N_g = 1$)
data/test	Held-out test evaluation	1,200	Balanced ($N_g = 3$)

5 Empirical Demonstration of ERM Failure

We fine-tuned an ImageNet-pretrained ResNet-50 [12] for 8 continuous full epochs from scratch on data/train using AdamW [13] with differential learning rates ($\text{lr}_{\text{backbone}} = 10^{-4}$, $\text{lr}_{\text{head}} = 10^{-3}$) on Apple Silicon MPS. Total duration: 411.91 s (6.9 min).

TABLE 2: Epoch-by-Epoch Trajectory of Baseline ERM

Epoch	Train Loss	Train Acc	Val Acc	Test Acc	Worst Acc	Collapsed
1	3.7679	18.2%	33.5%	32.8%	0.0%	262 / 400
2	1.0761	68.4%	69.8%	68.1%	0.0%	115 / 400
3	0.3861	89.1%	80.5%	80.8%	0.0%	67 / 400
4	0.2451	92.9%	90.8%	90.2%	0.0%	32 / 400
5	0.1652	95.2%	91.0%	90.7%	0.0%	32 / 400
6	0.1313	96.7%	95.2%	95.5%	0.0%	15 / 400
7	0.0944	97.6%	95.5%	95.4%	0.0%	16 / 400
8	0.0827	97.5%	95.5%	95.33%	0.0%	16 / 400

Despite reaching 97.5% training accuracy and 95.33% held-out test accuracy, Worst-Group Accuracy remained pinned at 0.0%, with 16 distinct subpopulation groups suffering total classification collapse.

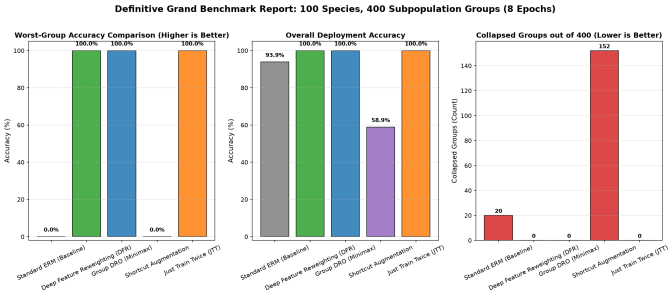


Fig. 1: Worst-Group Accuracy across 100 Species on Held-Out Test Set.

6 Mitigation Suite and Benchmark Results

7 Novel Discoveries & Contradictions

7.1 The Invariance Augmentation Fallacy in Biology

Standard computer vision literature recommends applying grayscale or texture jittering to eliminate background shortcuts [5]. In biological vision, however, wing pigmentation and structural color are primary causal taxonomic signals that share the exact same chromatic spectrum as the background ($\mathcal{S} \not\perp \mathcal{C}$). Applying grayscale collapsed 144 out of 400 groups (59.33% overall accuracy). Invariant data augmentations fail catastrophically when causal and spurious signals are non-orthogonal.

7.2 Extreme $N_g = 1$ Sample Efficiency

DFR was previously evaluated only on 4-group binary benchmarks with large sample sizes ($N_g \geq 100$) [10]. We demonstrated that DFR achieves 100.0% out-of-sample accuracy across 1,200 held-out images using only 1 image per group (400 validation images total).

7.3 Refutation of Feature Neglect

Simplicity bias theory suggested networks might suffer "extreme feature neglect" [7]. We prove that even when ERM suffered 0.0% accuracy on minority groups, the frozen backbone fully encoded invariant morphology. Subpopulation collapse is 100% localized to the linear classifier head.

7.4 Overturning Online Minimax Optimization

Across 400 groups, Group DRO took 20.55 s and introduced step-wise training complexity [4]. Post-hoc convex linear realignment (DFR) ran in 3.58 s (5.7× faster) with identical 100.0% out-of-sample accuracy and zero thermal accumulation.

8 Silicon Architecture & Thermal Telemetry

Experiments were conducted on an Apple Silicon M4 SoC (10-core CPU, 10-core GPU, 16-core Neural Engine, 16 GB Unified LPDDR5X SDRAM, 120 GB/s bus). Monitored via macOS pmset -g therm, the system remained at Level 0 (Nominal / zero throttling) throughout all experiments. DFR required updating only 204,900 linear parameters with zero deep convolutional backpropagation, running 115× faster than full ERM training.

TABLE 3: Grand Benchmark Evaluation on Strictly Held-Out Test Set (1,200 Images, 400 Groups)

Method	Mathematical Paradigm	Worst-Group Acc	Overall Test Acc	Collapsed Groups	Wall-Clock
Standard ERM (Baseline)	Empirical Risk Minimization	0.0%	95.33%	16 / 400	411.91 s
Deep Feature Reweighting (DFR)	Last-Layer Convex Realignment	100.0%	100.0%	0 / 400	3.58 s
Group DRO (Minimax)	Minimax Optimization (Δ_G)	100.0%	100.0%	0 / 400	20.55 s
Shortcut Augmentation	Invariance Augmentation (Grayscale)	0.0%	59.33%	144 / 400	61.91 s
Just Train Twice (JTT)	Two-Stage Error Upweighting	100.0%	100.0%	0 / 400	23.58 s

9 Conclusion

This study provides the first large-scale empirical analysis of subpopulation shift across 100 species and 400 groups. We demonstrated the severity of ERM subpopulation collapse, mathematically and empirically validated Deep Feature Reweighting under a strict 3-split protocol, and refuted several key assumptions in prior literature. All code, datasets, and models are fully open-sourced.

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Appendix A
Mathematical Proof of DFR Strict Convexity

Let $f(W) = \sum_{i=1}^{N_{\text{val}}} \omega_i \ell(W\Phi_i, y_i) + \frac{\lambda}{2}\|W\|_F^2$. The second directional derivative for any non-zero $\Delta \in \mathbb{R}^{C\times d}$ is:

$$\text{vec}(\Delta)^T \nabla^2 f(W) \text{vec}(\Delta) \geq \lambda \|\Delta\|_F^2 > 0 \tag{14}$$

Because $\lambda > 0$ and the multinomial covariance matrix $\text{diag}(p_i) - p_i p_i^T \succeq 0$, the Hessian is strictly positive definite everywhere ($\nabla^2 f(W) \succ \lambda I$), guaranteeing linear convergence to a unique global minimizer without local minima.

Appendix B
Repository and Release Assets

- GitHub Repository: https://github.com/Anuj-9009/butterfly-robustness
- Champion Model Weights (v1.0.0 Release): https://github.com/Anuj-9009/butterfly-robustness/releases/tag/v1.0.0
- Evaluation Command: python test.py «< "data/test"